Topic-wise DPP for NEET Aspirants

By Tamanna Chaudhary

Most Scoring Topic XIV - Moment of inertia

Chapter – Rotational Mechanics

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1. Moment of inertia of a disc about an axis which is tangent and parallel to its plane is I. Then the moment of inertia of disc about a tangent, but perpendicular to its plane will be

   a) $\frac{3I}{4}$ b) $\frac{5I}{6}$ c) $\frac{3I}{2}$ d) $\frac{6I}{5}$

2. Moment of inertia of big drop in I. If 8 droplets are formed from big drop, then moment of inertia of small droplet is

   a) $\frac{I}{32}$ b) $\frac{I}{16}$ c) $\frac{I}{8}$ d) $\frac{I}{4}$

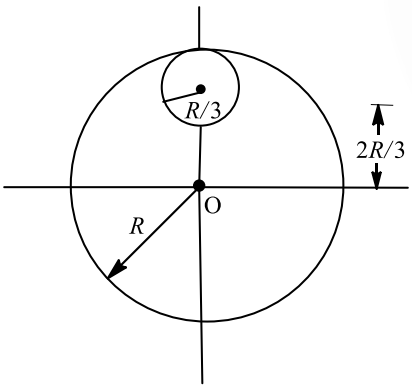
3. A spherical shell has mass M and radius R. Moment of inertia about its diameter will be

   a) $\frac{2}{5}MR^2$ b) $\frac{2}{3}MR^2$ c) $\frac{1}{2}MR^2$ d) MR^2

4. The moment of inertia of a body does not depend upon

   a) The angular velocity of the body b) The mass of the body
   c) The distribution of mass in the body d) The axis of rotation of the body

5. From a circular disc of radius R and mass $9\,M$, a small disc of radius $R/3$ is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is



   a) $4\,MR^2$ b) $\frac{40}{9}\,MR^2$ c) $10\,MR^2$ d) $\frac{37}{9}\,MR^2$

6. A wheel of mass $10\,kg$ has a moment of inertia of $160\,kg\text{-}m^2$ about its own axis, the radius of gyration will be

   a) $10\,m$ b) $8\,m$ c) $6\,m$ d) $4\,m$

7. Moment of inertia of a thin circular disc of mass M and radius R about any diameter is

   a) $\frac{MR^2}{4}$ b) $\frac{MR^2}{2}$ c) MR^2 d) $2MR^2$

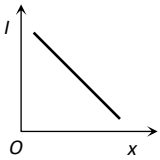
8. Moment of inertia of ring about its diameter is I. Then, moment of inertia about an axis passing through center perpendicular to its plane is

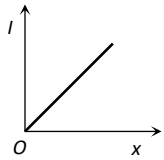
   a) $2I$ b) $\frac{I}{2}$ c) $\frac{3}{2}I$ d) I

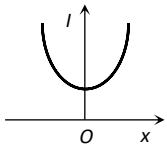
9. The moment of inertia of a rod about an axis through its center and perpendicular to it is $\frac{1}{2}ML^2$ (where M is the mass and L is the length of the rod). The rod is bent in the middle so that the two halves makes an angular of 60^0. The same axis would be

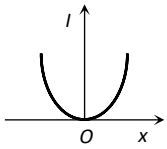
a) $\frac{1}{48}ML^2$ b) $\frac{1}{12}ML^2$ c) $\frac{1}{24}ML^2$ d) $\frac{ML^2}{8\sqrt{3}}$

10. According to the theorem of parallel axes $I = I_{cm} + Mx^2$, the graph between I and x will be

a) 

b) 

c) 

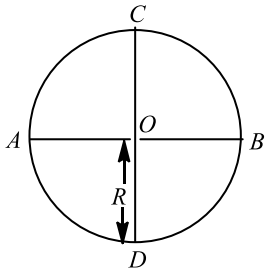
d) 

NEWTON'S APPLE

: HINTS AND SOLUTIONS :

1 **(d)**

The moment of inertia of the disc about an axis parallel to its plane is



$$I_t = I_d + MR^2$$

$$\Rightarrow \quad I = \frac{1}{4}MR^2 + MR^2$$

$$= \frac{5}{4}MR^2$$

or $\quad MR^2 = \frac{4I}{5}$

Now, moment of inertia about a tangent perpendicular to its plane is

$$I' = \frac{3}{2}MR^2 = \frac{3}{2} \times \frac{4}{5}I = \frac{6}{5}I$$

2 **(a)**

Moment of inertia of big drop is $I = \frac{2}{5}MR^2$. When small droplets are formed from big drop volume of liquid remain same

$$n\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\Rightarrow \quad n^{1/3}r = R$$

as $\quad n = 8$

$$\Rightarrow \quad r = \frac{R}{2}$$

Mass of each small droplet $= \frac{M}{8}$

$\therefore$ Moment of inertia of each small droplet

$$= \frac{2}{5}\left[\frac{M}{8}\right]\left[\frac{R}{2}\right]^2$$

$$= \frac{1}{32}\left[\frac{2}{5}MR^2\right] = \frac{I}{32}$$

3 **(b)**

4 **(a)**

5 **(a)**

$I_{\text{remaining}} = I_{\text{whole}} - I_{\text{removed}}$

or $\quad I = \frac{1}{2}(9M)(R^2) - \left[\frac{1}{2}m\left(\frac{R}{3}\right)^2 + \frac{1}{2}m\left(\frac{2R}{3}\right)^2\right]$

...(i)

Here, $\quad m = \frac{9M}{\pi R^2} \times \pi\left(\frac{R}{3}\right)^2 = M$

Substituting in Eq. (i), we have

$$I = 4MR^2$$

6 **(d)**

$$I = MK^2 = 160 \Rightarrow K^2 = \frac{160}{M} = \frac{160}{10} = 16 \Rightarrow K = 4metre$$

7 **(a)**

8 **(a)**

By the theorem of perpendicular axes, the moment of inertia about the central axis I_C, will be equal to the sum of its moments of inertia about two mutually perpendicular diameters lying in its plane.

Thus, $\quad I_d = I = \frac{1}{2}MR^2$

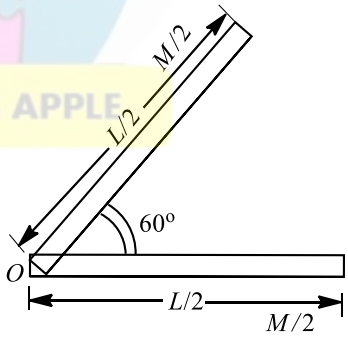
$\therefore \quad I_C = I + I$

$$= \frac{1}{2}MR^2 + \frac{1}{2}MR^2$$

$$= I + I = 2I$$

9 **(b)**

Since, rod is bent at the middle, so each part of it will have same length $\left(\frac{L}{2}\right)$ and mass $\left(\frac{M}{2}\right)$ as shown.



Moment of inertia of each part through its one end

$$= \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2$$

Hence, net moment of inertia through its middle point O is

$$I = \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2 + \frac{1}{3}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2$$

$$= \frac{1}{3}\left[\frac{ML^2}{8} + \frac{ML^2}{8}\right] = \frac{ML^2}{12}$$

10 **(c)**

Graph should be parabola symmetric to I-axis, but it should not pass from origin because there is a constant value I_{cm} is present for $x = 0$

Topic-wise DPP for NEET Aspirants

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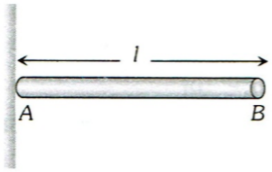
Most Scoring Topic XV - Torque

Chapter – Rotational Mechanics

1. A uniform rod AB of length l and mass m is free to rotate about point A. The rod is released from rest in horizontal position. Given that the moment of inertia of the rod about A is $\frac{ml^2}{3}$ the initial angular acceleration of the rod will be



a) $\frac{2g}{3l}$ b) $mg\frac{l}{2}$ c) $\frac{3}{2}gl$ d) $\frac{3g}{2l}$

2. A 10 kg body hangs at rest from a rope wrapped around a cylinder 0.2 m in diameter. The torque applied about the horizontal axis of the cylinder is

a) 98 N-m b) 19.6 N-m c) 196 N-m d) 9.8 N-m

3. A door 1.6 m wide requires a force of 1 N to be applied at the free end to open or close it. The force that is required at a point 0.4 m distance from the hinges for opening or closing the door is

a) 1.2 N b) 3.6 N c) 2.4 N d) 4 N

4. A uniform rod of length L and mass 1.8 kg is made to rest on two measuring scale at its two ends. A uniform block of mass 2.7 kg is placed on the rod at a distance $L/4$ from the left end. The force experienced by the measuring scale on the right end is

a) 18 N b) 27 N c) 29 N d) 45 N

5. Turning effect is produced by

a) Tangential component of force b) Radial component of force

c) Transverse component of force d) None of these

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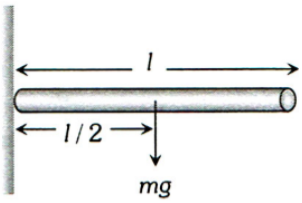
1 **(d)**

Weight of the rod will produce the torque

$\tau = I\alpha \Rightarrow mg \times \frac{l}{2} = \frac{ml^2}{3} \times \alpha$



Angular acceleration

$\alpha = \frac{3g}{2l}$

2 **(d)**

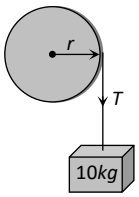
$\tau = r \times F$

$= r \times T$

$= r \times m \times g$

$= 0.1 \times 10 \times 9.8$

$= 9.8\ N\text{-}m$



3 **(d)**

Here, torque $\tau = 1.6 \times 1 = 1.6$ Nm

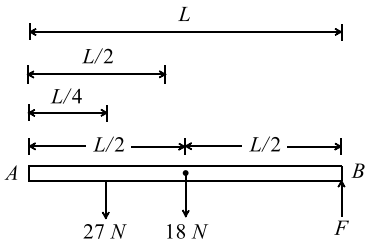
So, when $d = 0.4$ m,

$F = \frac{\tau}{d} = \frac{1.6}{0.4} = 4$ N

4 **(a)**

Mass of a rod, $m = 1.8kg$

$\therefore$ Weight of a rod, $W = mg = 1.8kg \times 10ms^{-2} = 18\ N$



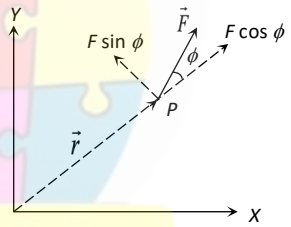
As the rod is uniform, therefore weight of the rod is acting at its midpoint

Taking moments about A,

$27 \times \frac{L}{4} + 18 \times \frac{L}{2} = F \times L$

$\Rightarrow FL = \frac{L}{4}[27 + 36] = \frac{63L}{4} \Rightarrow F = \frac{63}{4} = 16N$

5 **(c)**



$\vec{\tau} = \vec{r} \times \vec{F} = r\,F \sin\phi \ \therefore \tau = rF \sin\phi$

$F \sin\phi =$ transverse component of force

$F \cos\phi =$ radial component of force

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